



Near transfer to an unrelated *N*-back task mediates the effect of *N*-back working memory training on matrix reasoning

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The extent to which working memory training improves performance on untrained tasks is highly controversial. Here we address this controversy by testing the hypothesis that far transfer may depend on near transfer using mediation models in three separate randomized controlled trials (RCTs). In all three RCTs, totalling 460 individuals, performance on untrained *N*-back tasks (near transfer) mediated transfer to Matrix Reasoning (representing far transfer) despite the lack of an intervention effect in RCTs 2 and 3. Untrained *N*-back performance also mediated transfer to a working memory composite, which showed a significant intervention effect (RCT 3). These findings support a model of *N*-back training in which transfer to untrained *N*-back tasks gates further transfer (at least in the case of working memory at the construct level) and Matrix Reasoning. This model can help adjudicate between the many studies and meta-analyses of working memory training that have provided mixed results but have not examined the relationship between near and far transfer on an individual-differences level.

The potential that working memory (WM) training can transfer to untrained tasks that differ substantially from the training task ('far transfer') has generated substantial excitement with regard to better understanding the causal relationship between WM and related cognitive processes^{1–6}, as well as to applied domains where such training can benefit health and well-being^{7–10}. In particular, the effectiveness of WM training to improve fluid intelligence has been a controversial and highly debated topic, with some meta-analyses showing a small but significant positive effect on fluid intelligence^{11–15}, while others argue that there is no evidence of generalization to fluid intelligence after training^{16,17}. A difficulty in reconciling across different studies is that individual differences in performance and learning are ubiquitous in this literature and impact the average benefits¹⁸. Furthermore, training procedures and outcome measures differ greatly across studies, making meaningful comparisons challenging¹⁹.

One of the most common tasks used in these WM training studies is the *N*-back, an updating task that correlates significantly with measures of higher-level cognition, such as Matrix Reasoning^{20–24}. While there is little doubt that *N*-back training leads to improved performance on untrained *N*-back tasks^{13,25–27}, the extent to which this reflects the generation of performance strategies rather than improvement in monitoring and updating of WM content, or improved inhibition of items that are no longer relevant, remains unknown²⁵. If we assume the former, we have little reason to expect post-test improvement on tasks that are not of the same structure as the *N*-back, such as Matrix Reasoning. However, if *N*-back training leads to enhanced efficiency of updating and/or inhibitory control (IC) processes, such far transfer effects would be plausible. From a theoretical perspective, 'near' transfer is thought to occur when the training task and outcome measure share many overlapping features and presumably recruit similar processes, whereas 'far' transfer refers to improved performance on outcome measures that share fewer features with the training task. While the boundaries between

near and far transfer are not clear and probably fall on a continuum²⁸, several meta-analyses of WM training studies have demonstrated that near transfer effects are more readily observable and show larger effect sizes than far transfer effects^{13,14}. Likewise, neuroimaging research suggests that the frontostriatal system mediates near transfer effects, whereas the dorso- and ventrolateral prefrontal cortices seem to employ domain-general networks that facilitate learning more widely⁵.

Here we go beyond merely examining the effects of *N*-back training on Matrix Reasoning to attempt to understand how these effects occur in the first place²⁹. We use mediation models because they represent a powerful tool for making theoretical predictions about causal mechanisms when applied to experimental data³⁰. Specifically, we first present the results of a Retrospective study in which mediation models were used to evaluate whether the effects of *N*-back training on Matrix Reasoning are transmitted via changes in an untrained variant of the *N*-back. In addition, we used an IC task as a control measure for our analyses. Even though improvement in the training task itself may also serve as a useful mediator³¹, this information is not available for control groups, nor is it necessarily indicative of near transfer. We hypothesized that *N*-back training groups would improve on Matrix Reasoning (far transfer) compared with no-contact control groups, and furthermore that this effect would be mediated by post-test performance on an untrained *N*-back task (near transfer). This model enabled us to account for individual differences in responsiveness to WM training and subsequent transfer to untrained tasks, and to demonstrate that transfer to Matrix Reasoning tasks is completely mediated by near transfer in two independent randomized control trials (RCTs) collected over two academic years. Since demand characteristics during training and potential placebo effects were not controlled for in the Retrospective study, we further report the results of a double-blind, pre-registered Replication study with an active control group in which we replicate the finding that performance on

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untrained *N*-back tasks mediates transfer to Matrix Reasoning and extend the discussion to other tasks and domains.

Results

Retrospective study (RCT 1 and RCT 2). Conventional general linear model analyses test for differences in means of different groups on one independent variable and therefore cannot be used to test our theoretical model in which a given training intervention leads to far transfer completely or partially through near transfer gains. Instead, we use mediation models with the following hypothesized sequence of events: random group assignment (*N*-back training or control group) affects untrained *N*-back performance, which subsequently affects Matrix Reasoning performance. As a control analysis, performance on the Countermanding task (which also showed some training-related benefits) is entered as a parallel mediator to pit competing mechanisms against each other. We first evaluate the Retrospective study as a whole sample and then apply parallel two-mediator models separately for each RCT.

Whole sample. The linear regression results shown in Table 1 provide evidence that training group was a statistically significant predictor of post-test performance on Matrix Reasoning, even after controlling for pre-test performance on Matrix Reasoning, *N*-back and Countermanding. All variables except training group (0 = no-contact control, 1 = *N*-back training) were *z*-transformed to account for study-specific differences in unit of measurement.

After post-test *N*-back and Countermanding were added as parallel mediators (Fig. 1a), the total effect (TE) of training group on post-test Matrix Reasoning performance remained statistically significant (TE = 0.23; s.e. = 0.10; $p = 0.02$); the 95% confidence interval (CI) of this effect was [0.03, 0.43], with a partially standardized effect size (ES_{ps}) of 0.21, and a coefficient of determination (R^2) of 0.50. Training group was a statistically significant predictor of performance on post-test *N*-back ($B = 1.05$; $p < 0.001$; 95% CI [0.80, 1.31]), and post-test *N*-back was in turn a statistically significant predictor of performance on post-test Matrix Reasoning ($B = 0.18$; $p < 0.001$; 95% CI [0.09, 0.28]). A bootstrap CI for the indirect effect (IE) based on 5,000 bootstrap samples did not include zero (IE = 0.19; s.e. = 0.05; 95% CI [0.09, 0.31]), providing evidence for statistically significant mediation via post-test *N*-back. While training group was a statistically significant predictor of post-test reaction time (RT) performance on the Countermanding task ($B = -0.25$; $p = 0.01$; 95% CI [-0.46, -0.05]), we found no statistically significant evidence that post-test Countermanding performance predicted post-test Matrix Reasoning ($B = 0.07$; $p = 0.214$; 95% CI [-0.04, 0.19]) nor any evidence of a statistically significant IE of training group on post-test Matrix Reasoning with Countermanding performance as a mediator (IE = -0.02; s.e. = 0.02; 95% CI [-0.06, 0.01]). Moreover, the two IEs, via post-test *N*-back and via post-test Countermanding, were statistically significantly different from each other (IE = 0.21; s.e. = 0.06; 95% CI [0.09, 0.34]). These results suggest that the causal effect of training group (no-contact control or *N*-back training) on Matrix Reasoning is mediated by the extent to which participants improve on near transfer. To determine the reproducibility of these results, the same analyses were conducted on the two independent RCTs.

RCT 1. In RCT 1, the linear regression results revealed that training group was a statistically significant predictor of post-test Matrix Reasoning ($B = 0.05$; $p = 0.048$; 95% CI [0.00, 0.10]; $R^2 = 0.50$) after controlling for pre-test Matrix Reasoning, *N*-back and Countermanding. In the mediation analysis (Fig. 1b), the TE of training group on Matrix Reasoning was also statistically significant (TE = 0.40; s.e. = 0.20; $p = 0.048$; 95% CI [0.003, 0.79]; $ES_{ps} = 0.35$; $R^2 = 0.50$). Training group predicted performance on post-test *N*-back ($B = 1.07$; $p < 0.001$; 95% CI [0.50, 1.63]), which in turn predicted performance on post-test Matrix Reasoning ($B = 0.17$;

Table 1 | Linear model of predictors of post-test performance on Matrix Reasoning

	<i>B</i>	s.e. of <i>B</i>	β	<i>p</i>
Retrospective study (RCT 1 and 2 combined)				
Constant	0.12 (-0.04 to 0.28)	0.08		0.140
Pre-test Matrix Reasoning	0.66 (0.56 to 0.76)	0.05	0.60	<0.001
Pre-test <i>N</i> -back	0.09 (-0.02 to 0.19)	0.05	0.08	0.108
Pre-test Countermanding	-0.17 (-0.27 to -0.06)	0.05	-0.15	0.002
Training group	0.23 (0.03 to 0.43)	0.10	0.10	0.022
Replication study (RCT 3)				
Constant	0.25 (0.08 to 0.41)	0.09		0.01
Pre-test Matrix Reasoning	0.76 (0.65 to 0.88)	0.06	0.71	<0.001
Pre-test <i>N</i> -back	0.10 (0.002 to 0.19)	0.05	0.09	0.07
Pre-test Countermanding	-0.15 (-0.32 to 0.01)	0.09	-0.11	0.07
Training group	0.07 (-0.12 to 0.29)	0.10	0.03	0.50

Standardized and unstandardized regression coefficients are reported as *B* and β , respectively. The 95% bias-corrected and accelerated CIs are reported in parentheses. The CIs and standard errors are based on 1,000 bootstrap samples. Retrospective study, $R^2 = 0.60$; Replication study, $R^2 = 0.51$.

$p = 0.023$; 95% CI [0.02, 0.32]). A bootstrap CI for the specific IE did not include zero (IE = 0.19; s.e. = 0.09; 95% CI [0.03, 0.40]), providing evidence for statistically significant mediation. In contrast, there was no statistically significant evidence that performance on the Countermanding task mediated the relationship between training group and Matrix Reasoning (IE = -0.02; s.e. = 0.04; 95% CI [-0.12, 0.05]). The two IEs differed significantly from each other (IE = 0.20; s.e. = 0.11; 95% CI [0.01, 0.46]).

RCT 2. In RCT 2, the linear regression results provided no statistically significant evidence that training group predicted post-test Matrix Reasoning ($B = 0.03$; $p = 0.15$; 95% CI [-0.01, 0.07]; $R^2 = 0.51$) after controlling for pre-test Matrix Reasoning, *N*-back and Countermanding. Likewise, in the mediation analysis (Fig. 1c), there was no evidence of a statistically significant TE of training group on Matrix Reasoning (TE = 0.17; s.e. = 0.12; $p = 0.15$; 95% CI [-0.06, 0.40]; $ES_{ps} = 0.16$; $R^2 = 0.51$). Nevertheless, training group was a statistically significant predictor of performance on post-test *N*-back ($B = 1.06$; $p < 0.001$; 95% CI [0.77, 1.34]), and post-test *N*-back was in turn a statistically significant predictor of performance on post-test Matrix Reasoning ($B = 0.20$; $p = 0.002$; 95% CI [0.07, 0.32]) after controlling for baseline performance on each test. At post-test, all three outcome measures correlated significantly with each other (Supplementary Table 1), yet only the IE through the *N*-back task was significant (IE = 0.21; s.e. = 0.07; 95% CI [0.07, 0.36]), whereas there was no evidence of a significant IE through the Countermanding task (IE = -0.02; s.e. = 0.02; 95% CI [-0.07, 0.02]). The two IEs differed significantly from each other (IE = 0.23; s.e. = 0.09; 95% CI [0.07, 0.41]).

Consider two participants from this RCT who were assigned to the same training condition and who showed a similar training gain of about two *N*-levels ($S102_{\text{gain}} = 2.25$; $S105_{\text{gain}} = 2.17$) yet showed very different transfer to Matrix Reasoning: S105 showed no change at post-test relative to pre-test in terms of accuracy, whereas S102 showed improved accuracy at post-test. Examining the untrained *N*-back results reveals that S105 did not show evidence of near transfer, which may explain the lack of transfer to

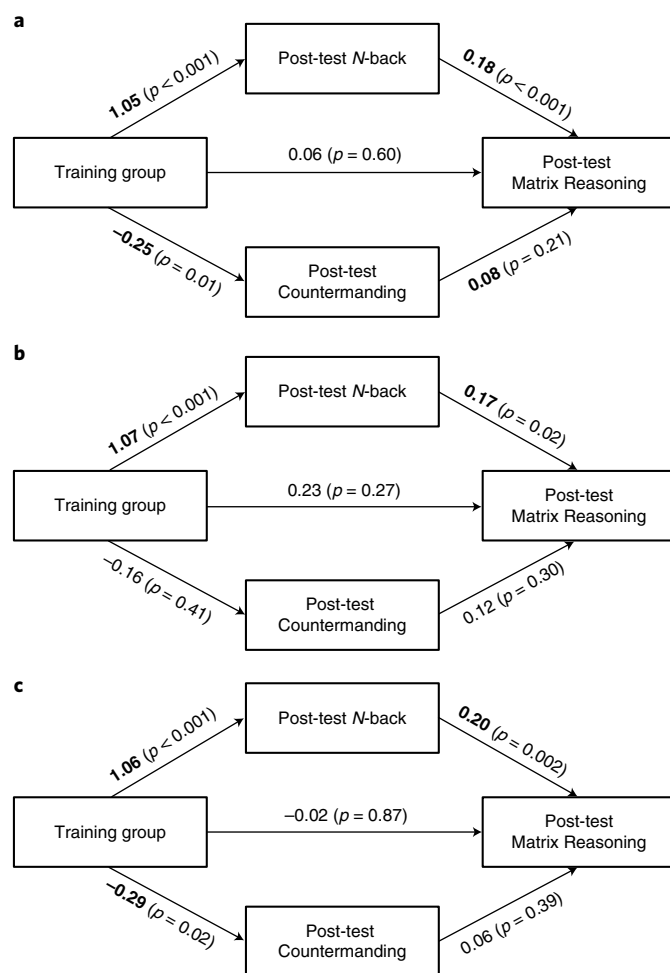


Fig. 1 | Parallel mediation models in the Retrospective study (RCT 1 and RCT 2). **a**, Retrospective study, whole sample ($n = 271$). **b**, Retrospective study, RCT 1 ($n = 94$). **c**, Retrospective study, RCT 2 ($n = 177$). In all models, the independent variable is training group (N -back training or no-contact control), the mediators are post-test N -back and Countermanding performance, and the dependent variable is post-test Matrix Reasoning performance. The path coefficients are standardized regression coefficients with p values reported in parentheses. Bold text indicates statistical significance. No corrections for multiple comparisons were made. The covariates (not displayed) were pre-test N -back, pre-test Countermanding and pre-test Matrix Reasoning.

Matrix Reasoning, whereas S102 showed near transfer. In this case, training gain has little predictive validity towards Matrix Reasoning outcomes. It remains to be determined why these participants showed such different patterns of results, but it is possible that individual differences in strategy sophistication³² and/or motivation³³ affected training outcomes.

Alternative mediation models. To test the hypothesis that N -back gains also mediate other aspects of transfer, we tested whether post-test N -back mediated the relationship between training group and post-test Countermanding performance (Extended Data Fig. 1). This was found in the whole sample ($IE = -0.23$; $s.e. = 0.09$; 95% CI $[-0.42, -0.08]$) and in RCT 2 ($IE = -0.28$; $s.e. = 0.12$; 95% CI $[-0.55, -0.08]$), but there was no statistically significant evidence that post-test N -back mediated the relationship between training group and post-test Countermanding in RCT 1 ($IE = -0.19$; $s.e. = 0.12$; 95% CI $[-0.46, 0.01]$).

Specific training effects and near transfer— N -back. Substantial individual differences in N -back performance were observed both as a function of training progression and at pre/post-test. For the N -back training groups, the mean N -level on the first day of training ranged from 2.00 to 6.44 in RCT 1 and from 1.12 to 4.84 in RCT 2. While these results suggest large individual differences in performance of all groups, training gain showed even greater variability: on the last day of training, the mean N -level ranged from 1.76 to 11.37 in RCT 1 and from 2.00 to 11.00 in RCT 2, with increasing variability as a function of training time (Extended Data Fig. 2). Larger training gain was associated with larger gains on the untrained N -back task in both RCTs (Supplementary Table 2).

No evidence of statistically significant baseline differences between N -back training and control groups was observed for the untrained N -back task (average accuracy across 2-back and 3-back levels; called Near Transfer hereafter) within each RCT (Supplementary Table 3). In RCT 1, pre-test accuracy on the Near Transfer task ranged from 0.42 to 0.93 in the no-contact control group and from 0.18 to 1.0 in the N -back training group. In RCT 2, pre-test accuracy on the Near Transfer task ranged from 0.22 to 0.82 for the no-contact control group and from 0.27 to 0.85 for the N -back training group. When we examined pre–post effect sizes (ES) across RCTs 1 and 2, the no-contact control groups showed little or no improvement at post-test relative to pre-test on Near Transfer, whereas the N -back training groups showed a substantial increase in accuracy on this task ($ES = 0.80$ to 0.92) (Table 2).

Far transfer effects. At pre-test, Near Transfer performance showed a statistically significant relationship with Countermanding and Matrix Reasoning performance in RCTs 1 and 2 (Supplementary Table 4; see Extended Data Fig. 3 for the scatter plots). Post-test scatter plots demonstrating relationships among the main outcome measures are shown in Extended Data Fig. 4 for RCT 1 and in Extended Data Fig. 5 for RCT 2. No evidence of statistically significant baseline differences between the N -back training and control groups was observed for Matrix Reasoning or Countermanding (Supplementary Table 3). The two N -back training groups showed small performance benefits on Matrix Reasoning ($ES = 0.43$ to 0.49), whereas the control groups showed negligible change in performance ($ES < 0.20$). Furthermore, the N -back training groups, but not the control groups, showed a decrease in average RT on incongruent trials of the Countermanding task, an index of IC, particularly in RCT 2 ($ES = 0.44$) (Table 2).

Given that near and far transfer effects were measured with single tasks, we recognize that the results might still reflect task-specific learning to some extent, and as such, we make no claims on the mental constructs of WM, response inhibition and fluid intelligence. Furthermore, due to the use of no-contact controls, it is possible that some of the performance advantage of the N -back training groups stems from non-specific characteristics of the training protocol, such as participant and/or experimenter expectancies and experimenter contact³⁴. Even though a recent meta-analysis found no meaningful performance difference between no-contact and active controls³⁵, we decided to conduct a pre-registered, double-blind Replication study using an active control group to control for these effects. Furthermore, we expanded the test battery to extend the analyses to other outcome measures and to be able to generate composite scores for each domain.

Pre-registered Replication study (RCT 3). In RCT 3, the analytical sample consisted of 189 participants: 128 participants completed 20 sessions of N -back training, whereas 61 participants completed 20 sessions of general knowledge training (active control). The N -back training group showed similar gains on Near Transfer ($ES = 1.52$), Matrix Reasoning ($ES = 0.46$) and Countermanding ($ES = 0.39$) (Table 3) as the training groups in the Retrospective study (RCT 1

Table 2 | Descriptive statistics, two-tailed Pearson correlation coefficients with corresponding Bayes factors, and Cohen's *d* effect sizes of pre- and post-test data in the Retrospective study (RCT 1 and RCT 2)

	Pre-test			Post-test			Correlation		Paired t-test	
	<i>n</i>	Mean	s.d.	<i>n</i>	Mean	s.d.	<i>r</i>	BF ₁₀	<i>d</i>	BF ₁₀
RCT 1										
Near Transfer										
N-back training	72	0.61	0.16	72	0.80	0.20	0.15	0.30	0.80	>100
No-contact control	22	0.61	0.15	22	0.62	0.22	0.52	4.58	0.06	0.23
Matrix Reasoning										
N-back training	72	0.72	0.13	72	0.76	0.14	0.73	>100	0.49	>100
No-contact control	22	0.71	0.11	22	0.71	0.16	0.51	3.97	−0.02	0.22
Countermanding										
N-back training	72	832.25	122.81	72	803.22	102.78	0.48	>100	0.25	1.05
No-contact control	22	842.87	112.43	22	827.22	136.90	0.58	11.70	0.14	0.27
RCT 2										
Near Transfer										
N-back training	109	0.59	0.15	109	0.78	0.18	0.31	20.29	0.96	>100
No-contact control	68	0.59	0.15	68	0.61	0.14	0.38	21.89	0.14	0.24
Matrix Reasoning										
N-back training	109	0.58	0.18	109	0.63	0.20	0.76	>100	0.43	>100
No-contact control	68	0.56	0.16	68	0.59	0.17	0.51	>100	0.19	0.44
Countermanding										
N-back training	109	745.42	120.43	109	710.46	109.01	0.77	>100	0.44	>100
No-contact control	68	756.61	129.13	68	757.37	179.04	0.68	>100	−0.01	0.13

Bayesian paired-samples *t*-tests with default priors were used to estimate whether there is a statistically significant difference in performance at post-test relative to pre-test for each measure. Reported are Bayes factors in favour of the alternative hypothesis over the null hypothesis (BF₁₀). Cohen's *d* was calculated as $(\text{Mean}_{\text{Post}} - \text{Mean}_{\text{Pre}}) / \sqrt{(s.d.^2_{\text{Pre}} + s.d.^2_{\text{Post}} - 2r_{\text{PrePost}} \times s.d._{\text{Pre}} \times s.d._{\text{Post}})}$. BF₁₀ > 10, indicating strong evidence for the alternative hypothesis, is marked with bold text.

and RCT 2). The basic assessment battery was expanded to include four additional WM tasks (Running Span, Sequencing, Simple Span and Complex Span), one additional measure of executive function (UCancellation) and three additional measures of spatial reasoning (ETS Surface Development, ETS Space Relations and DAT Form Board) to be able to construct composite scores in the three domains of interest. We first present a replication of the two-mediator parallel model, followed by models in which composite scores are entered as mediators and, finally, models in which a composite score represents the outcome variable.

On the basis of the linear regression results (Table 1), we found no statistically significant evidence that training group predicted post-test performance on Matrix Reasoning after controlling for pre-test performance on Matrix Reasoning, *N*-back and Countermanding. Likewise, in the mediation analysis (Fig. 2a), there was no evidence of a significant TE of training group on Matrix Reasoning (TE = 0.07; s.e. = 0.11; *p* = 0.51; 95% CI [−0.14, 0.28]; ES_{p.s.} = 0.07; *R*² = 0.60). Note that a simple association between independent and dependent variables is not necessary for mediation analysis³⁶. The parallel mediation model replicates the findings from the Retrospective study: training group was a statistically significant predictor of performance on post-test *N*-back (*B* = 1.34; *p* < 0.001; 95% CI [1.08, 1.60]), and post-test *N*-back was in turn a statistically significant predictor of performance on post-test Matrix Reasoning (*B* = 0.20; *p* < 0.001; 95% CI [0.08, 0.31]) after controlling for baseline performance on each test. A bootstrap CI for the specific IE through post-test *N*-back did not include zero (IE = 0.26; s.e. = 0.08; 95% CI [0.10, 0.43]), showing evidence for significant mediation. There was no statistically significant evidence that performance on the Countermanding task mediated the relationship

between training group and Matrix Reasoning (IE = 0.04; s.e. = 0.03; 95% CI [−0.004, 0.11]), which is in line with the results obtained in the Retrospective study. These results provide further evidence that transfer to untrained *N*-back tasks is a vital mediator through which the *N*-back training intervention affects improved performance on Matrix Reasoning. Moreover, post-test performance on the untrained *N*-back preceded performance on Matrix Reasoning by one day, hence satisfying the recommendation that the mediator must intervene between the predictor and the outcome³⁷. The same pattern of results was obtained when the two mediators were replaced with composite scores (Fig. 2b). Note that the sample size for this and subsequent analyses is smaller (*n* = 170) due to outliers removed on WM and IC tasks. Post-test WM Composite 1, which includes the *N*-back task, significantly mediated transfer to Matrix Reasoning (IE = 0.11; s.e. = 0.05; 95% CI [0.02, 0.22]), whereas we found no evidence of statistically significant mediation via the IC Composite (IE = 0.05; s.e. = 0.04; 95% CI [−0.003, 0.14]), thus replicating and extending the basic parallel model.

A question that arises is whether Matrix Reasoning is mediated solely by *N*-back or whether other WM measures contribute to the mediation as well. A multiple parallel mediation model (Fig. 3a) showed that post-test *N*-back was the only significant mediator of Matrix Reasoning performance (IE = 0.22; s.e. = 0.09; 95% CI [0.07, 0.41]). The IE via post-test *N*-back was significantly different from all other IEs (*N*-back versus Running Span: IE = 0.22; s.e. = 0.09; 95% CI [0.06, 0.41]; *N*-back versus Sequencing: IE = 0.22; s.e. = 0.09; 95% CI [0.06, 0.41]; *N*-back versus Simple Span: IE = 0.22; s.e. = 0.09; 95% CI [0.06, 0.41]; *N*-back versus Complex Span: IE = 0.21; s.e. = 0.09; 95% CI [0.04, 0.40]). Given these results, it is not surprising that WM Composite 2, which does not include *N*-back or

Table 3 | Descriptive statistics, two-tailed Pearson correlation coefficients with corresponding Bayes factors, and Cohen's *d* effect sizes of pre- and post-test data in the Replication study (RCT 3)

	Pre-test			Post-test			Correlation		Paired t-test	
	<i>n</i>	Mean	s.d.	<i>n</i>	Mean	s.d.	<i>r</i>	BF ₁₀	<i>d</i>	BF ₁₀
RCT 3										
Near Transfer										
N-back training	128	0.59	0.14	128	0.85	0.16	0.36	557.23	1.52	>100
Active control	61	0.60	0.18	61	0.66	0.17	0.63	>100	0.35	4.18
Matrix Reasoning										
N-back training	128	0.56	0.18	128	0.62	0.20	0.75	>100	0.46	>100
Active control	61	0.57	0.21	61	0.61	0.21	0.79	>100	0.36	5.06
Countermanding										
N-back training	128	784.49	126.81	128	750.19	113.70	0.74	>100	0.39	>100
Active control	61	775.01	15.09	61	782.83	18.20	0.53	>100	0.06	0.16
UCancellation										
N-back training	115	141.50	19.28	115	163.77	22.90	0.80	>100	1.61	>100
Active control	55	139.24	18.43	55	157.11	23.03	0.66	>100	1.02	>100
Running Span										
N-back training	115	23.38	9.33	115	26.62	10.72	0.09	0.17	0.24	2.23
Active control	55	24.18	10.25	55	26.36	11.09	0.35	4.77	0.18	0.34
Sequencing										
N-back training	115	8.58	3.28	115	9.44	3.74	0.53	>100	0.25	2.94
Active control	55	8.62	3.02	55	8.47	2.72	0.69	>100	0.06	0.16
Simple Span										
N-back training	115	52.89	15.08	115	54.89	15.28	0.49	>100	0.13	0.27
Active control	55	57.51	13.03	55	56.11	12.89	0.20	0.50	0.09	0.18
Complex Span										
N-back training	115	45.30	15.94	115	48.15	15.20	0.57	>100	0.20	0.87
Active control	55	45.96	13.26	55	45.96	15.19	0.45	47.68	0.00	0.15
ETS Surface Development										
N-back training	115	0.47	0.23	115	0.59	0.24	0.69	>100	0.68	>100
Active control	55	0.48	0.27	55	0.62	0.24	0.69	>100	0.72	>100
ETS Space Relations										
N-back training	115	0.66	0.20	115	0.71	0.19	0.57	>100	0.24	2.30
Active control	55	0.61	0.18	55	0.71	0.17	0.44	45.71	0.52	85.08
DAT Form Board										
N-back training	115	50.04	15.78	115	57.70	15.34	0.57	>100	0.53	>100
Active control	55	53.20	15.87	55	59.31	16.90	0.75	>100	0.53	95.36

Bayesian paired-samples *t*-tests with default priors were used to estimate whether there is a statistically significant difference in performance at post-test relative to pre-test for each measure. Reported are Bayes factors in favour of the alternative hypothesis over the null hypothesis (BF₁₀). Cohen's *d* was calculated as $(\text{Mean}_{\text{Post}} - \text{Mean}_{\text{Pre}}) / \sqrt{(s.d.^2_{\text{Pre}} + s.d.^2_{\text{Post}} - 2r_{\text{PrePost}} \times s.d._{\text{Pre}} \times s.d._{\text{Post}})}$. BF₁₀ > 10, indicating strong evidence for the alternative hypothesis, is marked with bold text.

Running Span, did not show a statistically significant mediation of the relationship between training group and transfer to Matrix Reasoning performance (IE = 0.006; s.e. = 0.004; 95% CI [−0.012, 0.015]), and neither did the IC Composite (IE = 0.011; s.e. = 0.007; 95% CI [−0.001, 0.027]) (Fig. 2c).

In the literature, the untrained *N*-back task is often seen as very near transfer, while structurally different WM tasks are classified as near transfer and non-WM tasks as far transfer¹³, implying a serial relationship among these task domains. While unlikely given the above results, a serial mediation model, in which the first mediator was post-test *N*-back and the second mediator was post-test WM Composite 2, was conducted to test this exploratory hypothesis

(Fig. 3b). As expected, the indirect specific effect via the untrained *N*-back was significant (training group → *N*-back → Matrix Reasoning: IE = 0.20; s.e. = 0.09; 95% CI [0.04, 0.39]), whereas we found no evidence of a significant IE when the model included WM Composite 2 as a mediator (training group → WM Composite 2 → Matrix Reasoning: IE = 0.01; s.e. = 0.02; 95% CI [−0.03, 0.05]; training group → *N*-back → WM Composite 2 → Matrix Reasoning: IE = 0.02; s.e. = 0.02; 95% CI [−0.01, 0.06]).

Since overriding dominant or prepotent responses is also required during *N*-back training (for example, when switching *N*-levels, the player must inhibit responses related to the previous level), and this may play a role in solving Matrix Reasoning tasks,

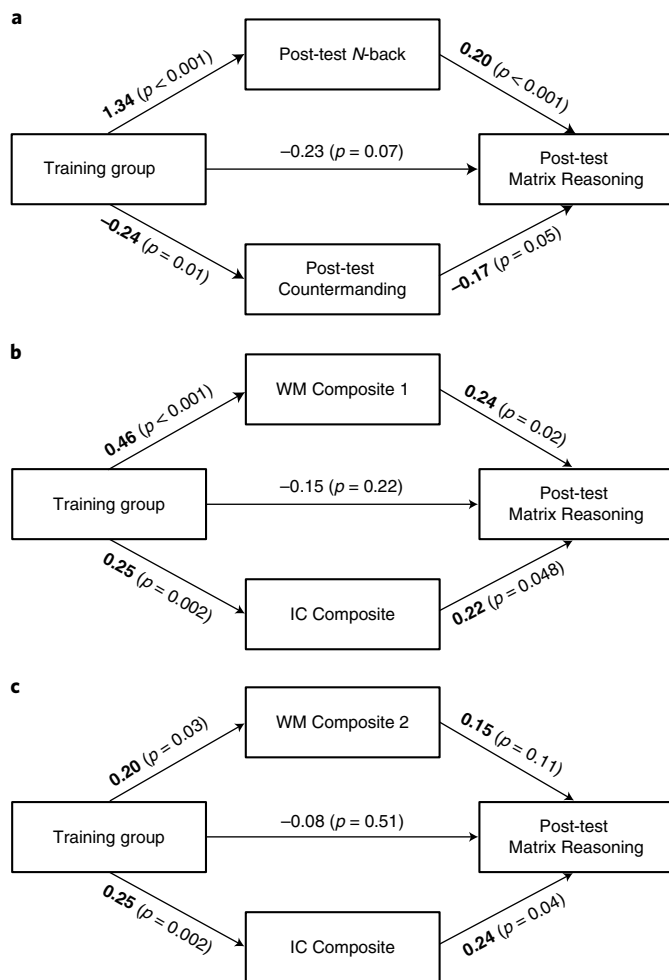


Fig. 2 | Parallel mediation models in the Replication study (RCT 3). **a**, RCT 3 ($n=189$). The mediators are post-test *N*-back and Countermanding. **b**, RCT 3—analytical subset ($n=170$). The mediators are WM Composite 1 (consisting of *N*-back, Sequencing, Simple Span and Complex Span) and IC Composite (consisting of Countermanding and UCancellation). **c**, RCT 3—analytical subset ($n=170$). The mediators are WM Composite 2 (consisting of Sequencing, Simple Span and Complex Span) and the IC Composite. In all models, the independent variable is training group (*N*-back training or active control), and the dependent variable is post-test performance on Matrix Reasoning, with pre-test data for all included measures entered as covariates. The path coefficients are standardized regression coefficients with p values reported in parentheses. Bold text indicates statistical significance. No corrections for multiple comparisons were made.

we replaced the second mediator (WM Composite 2) with the IC Composite (Fig. 3c). The results showed that only the path via *N*-back significantly mediated transfer to Matrix Reasoning (training group → *N*-back → Matrix Reasoning; IE = 0.21; s.e. = 0.09; 95% CI [0.05, 0.38]), whereas there was no evidence that the two IEs that included the IC Composite were significant (training group → IC Composite → Matrix Reasoning; IE = 0.04; s.e. = 0.03; 95% CI [-0.01, 0.12]; training group → *N*-back → IC Composite → Matrix Reasoning; IE = 0.02; s.e. = 0.02; 95% CI [-0.01, 0.05]). Overall, single-mediator models explained the data better than serial models. Note that the serial mediation models were not pre-registered and were added post hoc.

There is no statistically significant evidence that *N*-back training led to improved performance on other measures of visuospatial reasoning (ETS and DAT tests) relative to the active control group

(Table 3). Nevertheless, as planned during pre-registration, we ran a mediation model in which post-test *N*-back and Countermanding were entered as mediators and the outcome variable was a fluid intelligence composite (G_f), which consisted of Matrix Reasoning, ETS and DAT tests. While training group (*N*-back training or active control) predicted performance on post-test *N*-back ($B=1.22$; $p < 0.001$; 95% CI [0.95, 1.49]) and on post-test Countermanding ($B=-0.22$; $p=0.01$; 95% CI [-0.39, -0.05]), we found no evidence that *N*-back (IE = 0.03; s.e. = 0.05; 95% CI [-0.06, 0.13]) or Countermanding (IE = 0.03; s.e. = 0.02; 95% CI [0.0002, 0.08]) was a statistically significant predictor of performance on the G_f Composite after controlling for baseline performance on these tests. The same pattern of results was obtained when WM Composite 1 and the IC Composite were entered as mediators (Fig. 4a).

Alternative mediation models. Taking a step back, we investigated whether *N*-back-training-related changes are observed in WM tasks beyond the untrained *N*-back and what the mechanisms might be that support this process. Specifically, we tested whether performance on the untrained *N*-back mediates transfer to a WM composite consisting of sequencing and spatial span tasks, called WM Composite 2, while entering post-test Countermanding as a control mediator (Fig. 4b). The results showed that the IE through the untrained *N*-back was statistically significant (IE = 0.20; s.e. = 0.06; 95% CI [0.07, 0.30]), but there was no evidence of statistically significant mediation via Countermanding (IE = 0.01; s.e. = 0.02; 95% CI [-0.04, 0.05]). In addition, the TE of training group on post-test WM Composite 2 performance was statistically significant (TE = 0.21; s.e. = 0.09; $p=0.03$; 95% CI [0.02, 0.39]; $ES_{p.s.}=0.28$; $R^2=0.43$). These results suggest that the effectiveness of the *N*-back training intervention to improve WM skills depends on the extent to which an individual shows improvement on untrained *N*-back tasks. The same results were obtained when the Countermanding mediator was replaced with an IC Composite consisting of the Countermanding and UCancellation tasks (IE via IC = 0.03; s.e. = 0.02; 95% CI [-0.02, 0.08]).

We also ran a mediation model in which the IC Composite was entered as the dependent variable. There was no evidence that the indirect specific effects via the untrained *N*-back (IE = 0.05; s.e. = 0.05; 95% CI [-0.04, 0.15]) and WM Composite 2 (IE = 0.02; s.e. = 0.02; 95% CI [-0.01, 0.06]) were statistically significant. However, the TE of training group on IC was statistically significant (TE = 0.25; s.e. = 0.08; $p=0.002$; 95% CI [0.09 to 0.40]; $ES_{p.s.}=0.29$; $R^2=0.69$), suggesting that *N*-back training boosted IC performance relative to control training (Figs. 4c and 5c).

Specific training effects and near transfer. The *N*-back training group in RCT 3 showed similar training performance as in the Retrospective study, particularly with respect to RCT 2 (Extended Data Fig. 2). The mean *N*-level on the first day of training ranged from 2.00 to 3.94 and on the last from 2.00 to 14.33. Likewise, individual differences were observed in the Near Transfer task, with pre-test accuracy ranging from 0.22 to 0.95 in the active control group and from 0.23 to 0.95 in the *N*-back training group.

Far transfer effects. At pre-test, Near Transfer performance showed a statistically significant relationship with Countermanding and Matrix Reasoning performance (Supplementary Table 5; see Extended Data Fig. 6a–c for the scatter plots), which is in line with the findings reported for the Retrospective study. Post-test correlation statistics are shown in Supplementary Table 6, and scatter plots are shown in Extended Data Fig. 6d–f. No evidence of baseline differences between the *N*-back training and active control groups was observed for Matrix Reasoning or any of the other outcome measures (Supplementary Table 3). To confirm the results of the mediation analyses, Bayesian repeated-measures analyses of

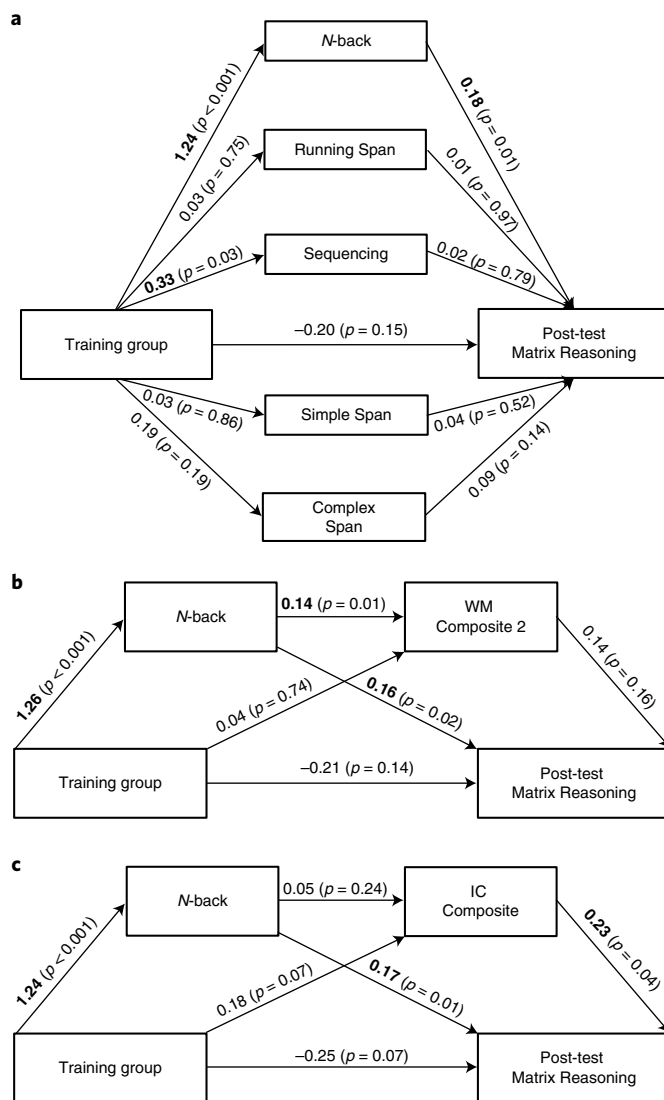


Fig. 3 | Alternative mediation models of transfer to Matrix Reasoning in the Replication study (RCT 3). **a**, Parallel multiple-mediator model. The mediators are post-test N-back, Running Span, Sequencing, Simple Span and Complex Span tasks. **b**, Serial mediator model in which the first mediator is post-test N-back and the second is WM Composite 2, consisting of the Sequencing, Simple Span and Complex Span tasks. **c**, Serial mediator model in which the first mediator is post-test N-back and the second is the IC Composite, consisting of the Countermanding and UCancellation tasks. In all models, the independent variable is training group (N-back training or active control), and the dependent variable is post-test performance on Matrix Reasoning, with pre-test data for all included measures entered as covariates. The path coefficients are standardized regression coefficients with p values reported in parentheses. Bold text indicates statistical significance. No corrections for multiple comparisons were made. RCT 3—analytical subset ($n = 170$).

variance with the within-participants factor time (pre or post) and the between-participants factor group (control or N-back training) were conducted on composite scores (not pre-registered). Inclusion Bayes factors across matched models (BF_{incl}) were used to evaluate evidence of time $\times$ group interactions. Statistically significant interactions in favour of the N-back training group were obtained for WM Composite 1 ($BF_{incl} > 100$), WM Composite 2 ($BF_{incl} = 2.46$) and the IC Composite ($BF_{incl} = 9.59$), but there was no evidence of a statistically significant interaction effect for the G_r

Composite ($BF_{incl} = 0.22$) (Fig. 5), which is in line with the results reported above.

Discussion

Previous studies have examined the relationship between training gain and far transfer, typically by comparing subgroups of participants who show low versus high training gain^{31,38,39}, however, this type of responder analysis can be problematic⁴⁰. Here we demonstrate that there are alternative ways of analysing transfer data that do not involve dichotomization and may reveal concealed associations and mechanisms to explain the effects of training. We obtained inconsistent results for the overall intervention effects on Matrix Reasoning across three RCTs; however, all RCTs, involving a total of 460 participants, showed that the extent of transfer to untrained N-back tasks determines transfer to Matrix Reasoning. This result may explain some of the variability seen in the literature that reports only sporadic transfer to these types of tasks. Transfer to untrained N-back tasks is often not reported; therefore, information is often missing about what participants learned, and the extent of learning on measures of transfer thus remains unknown. For example, in a meta-analysis of WM training studies¹³, 35 experiments measured fluid intelligence, 28 experiments assessed WM (including tasks that correlate poorly with the N-back²⁴) but only 9 experiments also included untrained N-back tasks. In another meta-analysis of N-back training studies, only 37% of N-back training studies reported using N-back tasks as an outcome measure¹⁹; thus, for the majority of studies, information is missing about what participants gained. If we do not know whether the minimal amount of near transfer occurred (for example, to a different stimulus set), then how can we gauge improvements in WM and measures beyond WM such as fluid intelligence or IC tasks?

The Retrospective WM training study using a no-contact control group confirmed our theoretical prediction that if N-back training improves performance on untrained N-back tasks, this will in turn lead to transfer to Matrix Reasoning, as demonstrated by a significant mediation effect and a significant total (intervention) effect. Note that the TE was demonstrated for the whole sample, but when examined separately for each of the two RCTs, it was demonstrated only for RCT 1. These results imply that N-back training does not merely enhance task-specific strategies but may lead to changes in underlying components of WM, such as the capacity of the focus of attention⁴¹ or updating⁴², which in turn can support performance on Matrix Reasoning problems⁴³. Understanding how this occurs is a challenge, which is constrained by the lack of agreement about why WM and measures of fluid intelligence, and in particular, Matrix Reasoning, correlate positively in the first place. One explanation suggests that WM is needed to maintain rules in mind to successfully solve Matrix Reasoning problems⁴⁴. However, this was not supported in a subsequent study; instead, the ability to manage proactive interference seemed to play a role in both constructs⁴⁵. Similarly, it has been shown that WM helps keep previously learned rule combinations from interfering when solving problems that require a new rule combination^{46,47}. However, since we found no evidence that IC mediated transfer to Matrix Reasoning, our results do not support interference-type accounts. Recent evidence also challenges these accounts, showing that the relationship between WM capacity and Matrix Reasoning can be explained by individuals' ability to maintain how they solved previous problems in memory⁴⁸. Our results are therefore in line with research showing that updating⁴² and executive attention⁴⁹ show strong relations with Matrix Reasoning performance. Finally, it has been argued that WM and fluid intelligence are complementary processes that promote complex cognition, wherein the first reflects maintenance and the other disengagement, and both serve a single top-down executive attention⁵⁰. Extending this to the neural level, it is plausible that transfer occurs if the training and transfer tasks engage overlapping processes and neural circuits^{26,51}.

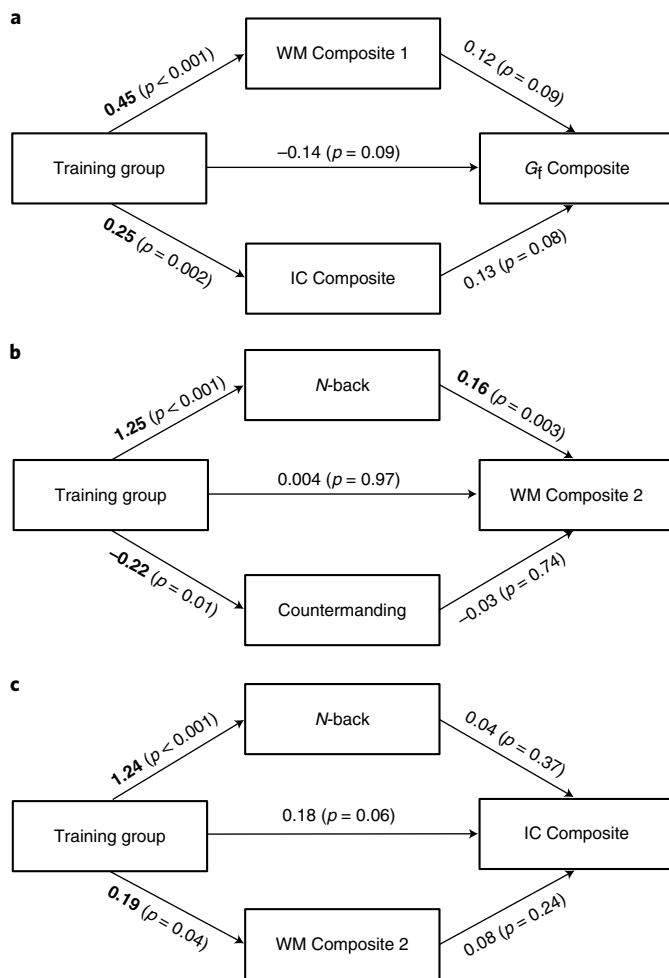


Fig. 4 | Alternative mediation models of transfer to composite measures in the Replication study (RCT 3). **a–c.** In each model, the independent variable is training group (*N*-back training or active control), and the dependent is *G*_f Composite (**a**), WM Composite 2 (**b**) or IC Composite (**c**), with pre-test measures entered as covariates. The path coefficients are standardized regression coefficients with *p* values reported in parentheses. Bold text indicates statistical significance. No corrections for multiple comparisons were made. RCT 3—analytical subset (*n* = 170).

To verify this finding using a more rigorous design, we pre-registered a double-blind Replication study and included an active control group, which was conducted online due to a ban on in-person research amid the COVID-19 pandemic. Furthermore, we investigated whether the mediation model replicates using composite scores of WM, IC and fluid intelligence as outcome measures. This study provided three key findings. First, we replicated the finding that transfer to Matrix Reasoning depends on the extent to which individuals show near transfer as measured by performance on untrained *N*-back tasks, even after controlling for participant and/or experimenter expectancies with the use of an active control group and full blinding procedures. Overall, this finding has now been observed in three independent RCTs that included a total of 460 individuals. While a significant total (intervention) effect on Matrix Reasoning was observed for the whole sample in the Retrospective study, this was not observed in the Replication study (RCT 3). Here, it is important to note that the mediated effect and the non-significant intervention effect each provide valuable information about how an intervention achieved (or failed to achieve) its effects; therefore, it is critical to discuss both types of

effects. There is evidence to suggest that in certain situations, mediated effects can be statistically significant in the absence of a TE. For example, the power of mediation is probably greater than the power of the test of the TE, because the mediator (the untrained *N*-back) is more closely related to the outcome measure (Matrix Reasoning) than the intervention is, and it was measured more closely in time with the outcome measure than the intervention⁵². In addition, it has been shown that the intervention can exert an IE on the outcome measure through a mediator in the absence of an association between these two variables, for example in the case of a crossover interaction⁵³.

It remains to be determined why some individuals show no or very little near transfer. It is possible that the individuals in our sample had low achievement motivation^{42,43}, showed little engagement with the task or applied inefficient strategies during training⁴⁴, resulting in poor WM training performance and, consequentially, limited near transfer. On the basis of subjective observation of the young-adult college student population that is represented in the sample, we note that they are at times sleep deprived, experience stress and anxiety, and adopt strategies of disengagement as the novelty of tasks wears off. Critically, the pre-registered Replication study was conducted remotely, with participants completing sessions in their home environments during social distancing amid the COVID-19 pandemic. It is now well documented that the pandemic was associated with widespread emotional and mental distress and increased risk for psychiatric illness^{54,55}. Elevated levels of COVID-19-related anxiety have also been associated with poorer WM performance as measured by *N*-back tasks⁵⁶. In our population, it was typical for students to have limited personal space and numerous home responsibilities and distractions, which may have increased the proportion of participants that were motivated to complete the study so that they would receive payment, but perhaps were not motivated or able to perform their best on the assigned tasks. This may explain the lack of an intervention effect on Matrix Reasoning in the Replication study (RCT 3). While these effects are clear from observation, we are still working on how to best characterize these factors objectively to more effectively disentangle participants who may ‘tune out’ from those who may be engaged but not learning. For now, the mediation analysis presented here is a first approach to addressing their contributions to the mean effects.

A second finding is that no WM measure other than the *N*-back task mediated transfer to Matrix Reasoning, even though post-test performance on all WM tasks except for Running Span correlated significantly with Matrix Reasoning performance (Supplementary Table 6). Span-based WM tasks might not share enough variance with Matrix Reasoning to explain individual differences in transfer, or the gain on these tasks might not be meaningful enough to have an observable effect on transfer.

A third finding is that *N*-back training led to small but significant improvements in a WM composite consisting of sequencing and span-based tasks and to enhanced IC performance relative to control training, as evidenced by statistically significant intervention effects. Untrained *N*-back performance mediated transfer to the WM composite, but there was no evidence of statistically significant mediation via the IC composite. Transfer to sequencing and/or complex WM tasks has been reported by some^{57–63} but not other *N*-back training studies^{58,64–70}. Likewise, there is no consensus regarding transfer to IC measures, with some WM training studies reporting training-related benefits^{59,60,71,72} but others failing to find evidence of transfer^{51,73}. Note that previous research often included Stroop tasks as an IC measure, whereas here we used a composite score based on speed of IC and a timed test of selective attention. Along with using a consistent set of measures across studies¹⁹, another solution that could bring some clarity is examining individual differences in training and transfer with mediation and moderation analyses. For example, the results of exploratory serial

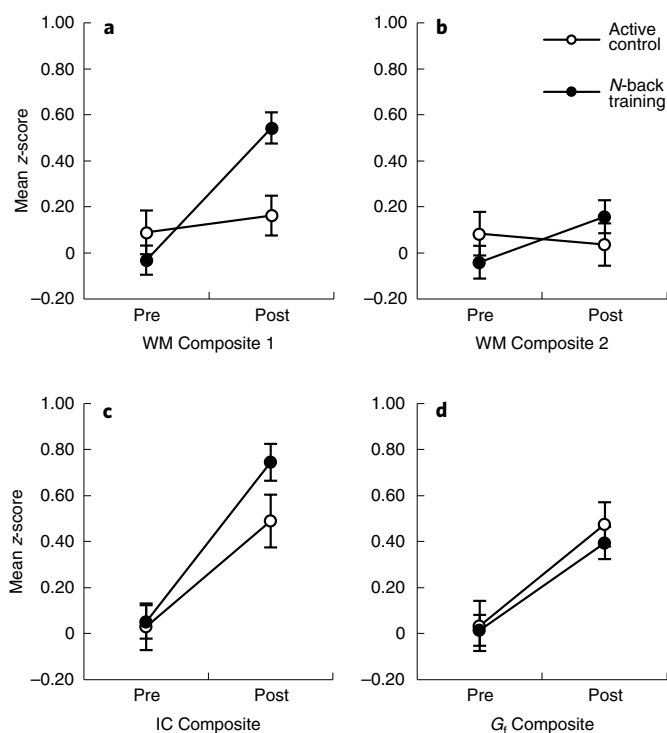


Fig. 5 | Composite z-scores at pre-test and post-test in the Replication study (RCT 3). **a**, WM Composite 1, consisting of the *N*-back, Sequencing, Simple Span and Complex Span. **b**, WM Composite 2, consisting of Sequencing, Simple Span and Complex Span. **c**, IC composite, consisting of Countermending and UCancellation. **d**, *G_f* Composite, consisting of UCMRT, ETS Surface Development, ETS Space Relations and DAT Form Board. The data are presented as mean values $\pm$ s.e.m., plotted separately for the *N*-back training (filled circles; $n=115$) and active control (empty circles; $n=55$) groups.

mediation showed that viewing the untrained *N*-back as very near transfer, which in turn affects transfer to other WM tasks or to an IC composite, and finally to Matrix Reasoning, does not explain the results. Instead, training-related improvement in IC could be seen as a separate form of (far) transfer, and the processes that underly this change seem to be somewhat independent.

There was no evidence of systematic change in a composite of *G_f* tasks as a function of WM training, and *N*-back performance did not seem to mediate transfer to this composite; therefore, we make no claims of transfer to fluid intelligence at the construct level. The Matrix Reasoning task used here (called UCMRT) has been shown to correlate highly with performance on Raven's Advanced Progressive Matrices⁷⁴; however, the correlations between UCMRT and the spatial visualization measures used here were much lower, especially for Surface Relations and Form Board (Supplementary Table 5), suggesting that the tasks share few commonalities. While UCMRT requires some spatial visualization, it is thought to tap mainly into abstract reasoning and logical problem solving. It is thus possible that UCMRT captures variance that is shared with the *N*-back but is different from the spatial visualization tasks used here. Of note, the Replication study showed a significant intervention effect as well as significant mediation to the WM composite, and thus, a relevant contribution of the paper is that near transfer after WM training mediates far transfer, which is not challenged by the fact that the *G_f* Composite did not show an effect. We consider both the Matrix Reasoning task and the WM composite as far transfer because they consist of tasks that are substantially different from the training

task. These results are in line with the finding that training-induced changes in WM capacity in children explain considerable parts of the intervention effect of WM training on far transfer⁷⁵. Because we did not perform follow-up assessments, we do not know how long any of the transfer effects last; however, there is some evidence to suggest that such effects are long-term^{76,77}.

A limitation of the Replication study (RCT 3) is that due to the COVID-19 pandemic, we were not able to conduct it in the lab; therefore, we were not able to completely match the conditions of the Retrospective study (RCT 1 and RCT 2). The participants completed the study on their own devices at home and were instructed to find a quiet environment for each session; however, it is likely that more distractions occurred compared with a controlled laboratory setting. That said, in the Retrospective study, multiple participants completed assessment and training sessions in the same room; thus, even this environment wasn't distraction-free. The level of supervision was different across studies given that all training sessions were supervised in the Retrospective study and only the first training session was supervised in the Replication study (or more, if needed). Given that the training curves for RCT 2 and RCT 3 are highly similar, yet both show slightly lower average training performance than that for RCT 1 (Extended Data Fig. 2), it is possible that factors besides the extent of supervision, such as small changes in the game mechanic or adoptive algorithm implemented after RCT 1 had concluded, affected these results. Furthermore, a recent comparison of cognitive assessment and training studies conducted in the lab and remotely via videoconferencing revealed similar compliance and performance rates⁷⁸. Another limitation is the young age of the sample in all three RCTs. Therefore, the extent to which the current results can be generalized to the many studies examining the effects of cognitive training in populations with diverse cognitive needs remains to be clarified.

Overall, our findings challenge the notion that WM training leads to performance gains only in tasks that are closely related to the training⁷⁹ and indicate that performance in untrained *N*-back tasks provides information regarding the effectiveness of an *N*-back intervention and should therefore be included as an outcome measure in future work. It remains to be seen whether a similar association between untrained tasks and transfer applies to other types of cognitive training paradigms or other types of transfer measures, including those capturing real-world performance. Moreover, these results can inform studies examining neuroplastic changes associated with WM interventions, predicting weaker far transfer effects in individuals who show little to no near transfer, in terms of behaviour and related plastic neural changes. From a practical perspective, a near transfer test administered halfway or early in the WM training intervention could indicate the likelihood of far transfer. If participants struggle in near transfer tasks at this point, and far transfer therefore seems unlikely, pivoting to a different type of intervention or providing additional support in terms of effective strategy use⁸⁰ could be beneficial to study outcomes (but see ref. ⁸¹). This simple yet presently untested step could help maximize the effectiveness of WM training interventions.

In sum, it is not surprising that inter-individual differences in responsiveness to WM training interventions exist^{82,83}—in non-invasive brain stimulation studies^{84,85} and other intervention studies, such as those focusing on physical exercise or nutrition^{86–89}, 'non-responders' are routinely excluded from data analyses or treated differently. To fully understand the underlying mechanisms of WM training, future research should depart from the traditional approach that attempts to determine whether an effect is present or not, and instead focus on how, for whom and why these effects occur^{31,90}. With the rise of precision medicine⁹¹, investigating the efficacy of personalized cognitive training interventions^{92,93} will become increasingly important and ultimately maximize intervention outcomes.

Methods

Retrospective study (RCT 1 and RCT 2). *Participants.* Our sample consisted of 283 participants from the Universities of California, Riverside and Irvine, collected over two academic years: winter–spring 2017 ($n_{\text{RCT1}} = 102$) and fall 2017–spring 2018 ($n_{\text{RCT2}} = 181$). The participants were randomly assigned either to *N*-back training ($n = 191$) or to no-contact control ($n = 92$) and completed all sessions in laboratories at the respective universities. They were included in the analysis if they completed all pre- and post-assessments as well as at least eight out of ten training days (*N*-back training group). Note that unequal sample sizes reflect randomization to multiple training conditions and one no-contact condition. The sample demographics are reported in Supplementary Table 7. The sample was relatively diverse, with 31–40% of participants identifying as Hispanic or Latinx and 42–44% identifying as Asian (Supplementary Table 8). The participants provided informed consent and received US\$120 compensation for participation; all procedures were approved by the University of California, Riverside Institutional Review Board for Human Subjects Research.

Procedure. Training and testing were administered on tablet computers via publicly available software that was developed in-house (Recollect the Study⁹⁴), in combination with Qualtrics Software used to administer surveys and certain assessments (Qualtrics). The same software for testing and training was used in both RCTs. The participants completed pre- and post-tests with a ten-day gap (excluding weekends); in addition, the *N*-back training groups completed up to ten working days of training. Across both RCTs, a variety of assessment tasks were used, including measures of WM, selective attention, mathematical problem solving, decision making and convergent thinking; however, to maximize power, here we focus only on the results of the three tasks that were used by both RCTs and that did not undergo major changes from one year to the next. Many of the measures not included here were being continuously developed for the use of tablets and were included in the studies for validation purposes. Moreover, our goal was to test the specific hypothesis that a simple mediation model focusing on the *N*-back and Matrix Reasoning can explain a substantial amount of variance in the data.

Training tasks. *N-back training.* The participants trained on an *N*-back task (Recollect the Study⁹⁴) for ten working days, each day consisting of two 20-minute training sessions separated by a 10- to 20-minute break. The training parameters were varied in that they differed in terms of *N*-level progression, gamification and reinforcement, but since these are not relevant to the present hypothesis, the data were collapsed across *N*-back training conditions. To provide context, in gamified *N*-back training, the participants played a platform game called Recollect the Study (a video can be found here: <https://www.youtube.com/watch?v=zhgL8Oe42Yk>) in which they were tasked with collecting stimuli that matched those presented *N* items earlier. The stimuli consisted of simple coloured objects and were paired with unique sounds to support multisensory processing of said objects⁶³. In the non-gamified *N*-back training (Tapback, <https://www.youtube.com/watch?v=GMGiDnJ53RU>), a series of coloured circles was presented on a plain background, and the participant tapped on the circles that were the same colour as those presented *N* items earlier. In all training paradigms, the stimuli were presented for 2,500 ms (interstimulus interval (ISI), 500 ms), of which 30% were targets and another 30% lures (for example, items that occur *N* – 1 or *N* + 1 of the target position)⁹⁴. The session was broken into roughly two-minute training blocks with typically seven to ten blocks per training session. Auditory and visual feedback was presented on every trial, indicating correct or incorrect responses. Task difficulty was adaptively adjusted on the basis of performance either within or between blocks depending on the training condition. The outcome measure used here was average *N*-level per day (weighted across the number of trials within each block).

Near Transfer *N*-back. We used an untrained *N*-back task with two versions, featuring pictures of animals or vehicles. The versions were counterbalanced across individuals and sessions, except for 40% of the participants in RCT 1, who completed the same version twice due to a technical error. Mann–Whitney tests indicated that there were no significant performance differences at post-test as a function of test version (different versus the same) in the no-contact ($U = 40$, $z = -0.54$, $p = 0.88$) and *N*-back training ($U = 624$, $z = -1.01$, $p = 0.27$) groups. All participants completed a 1-back task, followed by 2- and 3-back tasks. Before each level started, there were ten trials of practice during which performance feedback was provided. In RCT 1, the test phase for each *N*-level consisted of 20 + *N* trials with six targets, whereas in RCT 2, each *N*-level consisted of 30 + *N* trials with nine targets (30% target and 30% lure rates in both RCTs). Progression to 4-back (and beyond) was allowed if the participant made no more than two errors on the previous level; otherwise, the task stopped at 3-back. Accuracy was calculated for each *N*-level as hits/(hits + misses + false alarms), allowing for comparison across RCTs. Accuracy at the 1-back level was very high (RCT 1: $\text{Mean}_{\text{pre}} = 0.88$, $\text{s.d.}_{\text{pre}} = 0.18$, $\text{Mean}_{\text{post}} = 0.94$, $\text{s.d.}_{\text{post}} = 0.14$; RCT 2: $\text{Mean}_{\text{pre}} = 0.90$, $\text{s.d.}_{\text{pre}} = 0.16$, $\text{Mean}_{\text{post}} = 0.95$, $\text{s.d.}_{\text{post}} = 0.11$), and only 7.4% and 4.8% of participants reached *N*-levels higher than 3 at pre-test, respectively; thus, we used average accuracy across the 2-back and 3-back levels as the dependent variable (at least 40 + *N* trials per participant).

Matrix Reasoning. Matrix Reasoning problems were used to assess non-verbal problem solving ability⁹⁵. Different versions of the test were used at pre-test and post-test, counterbalanced across participants. While the number of problems and certain design features differed in the two RCTs, the basic structure of the task remained the same. Namely, each problem consisted of a 3 × 3 matrix with the lower right entry missing, and the participants were asked to select one out of eight answer options that best completed the matrix. Prior to the test phase, the participants were presented with example problems and the opportunity to practise the task, during which feedback was provided. In RCT 1, the participants were required to solve 42 problems with a time limit of 20 minutes⁹⁵, whereas in RCT 2, the participants solved 23 problems with a time limit of 10 minutes (UCMRT)⁹⁴. The main outcome measure in both RCTs was the proportion of correctly solved problems; in RCT 1, there was a larger number of easy problems (that is, one- and two-relation problems) than used in RCT 2; hence, accuracy was higher for RCT 1 (Table 2).

Countermanding task. A modified Countermanding task was used to assess IC^{76,96}. A series of pictures appeared on either the left or the right of the screen, and the participants were instructed to quickly and accurately tap one of two buttons presented below. The task involved pressing the button on the same side of the screen in response to pictures of a dog (congruent trials) and the button on the opposite side in response to pictures of a monkey (incongruent trials). There were three blocks of trials, each preceded by a practice session: a block of 12 congruent trials, a block of 12 incongruent trials and a mixed block of 48 trials where congruent and incongruent trials were randomly intermixed. All RCTs showed high accuracy in the mixed block: 97% on average or higher at pre-test and post-test. The main dependent measure was RT for correct responses on incongruent trials of the mixed block (index of IC).

Outliers and exclusions. Six outliers were removed from the sample on the basis of performance on Near Transfer and Matrix Reasoning ($|z_{\text{accuracy}}| > 3$ at pre- or post-test separately for each RCT). In the *N*-back training group, six additional participants were removed due to a software error in the training program. The final analytical sample consisted of 271 participants, 181 in the *N*-back training group (RCT 1: $n = 72$; RCT 2: $n = 109$) and 90 in the no-contact control group (RCT 1: $n = 22$; RCT 2: $n = 68$).

Data analysis. JASP version 0.14.1 (ref. ⁹⁷) was used to run Bayesian paired-samples *t*-tests with default priors to estimate changes in performance as a function of time separately for each training condition and RCT (Tables 2 and 3). A value of BF_{10} between 1 and 3 is considered to be only anecdotal evidence for the alternative hypothesis, 3–10 is moderate evidence, 10–30 is strong evidence, 30–100 is very strong evidence and >100 is extreme evidence^{98,99}. All variables were *z*-transformed (on the basis of pre-test scores, separately for each RCT) prior to regression-based analyses to control for RCT effects. Linear regression was used to test the relative contributions of training group (*N*-back training or control), pre-test Matrix Reasoning scores, pre-test *N*-back and pre-test Countermanding to post-test performance on Matrix Reasoning. We checked for multicollinearity, heteroscedasticity and independent errors using a Durbin–Watson test¹⁰⁰ and found these to be in the appropriate ranges.

To determine whether the relationship between training group and Matrix Reasoning (Far Transfer) was mediated by Near Transfer, we conducted mediation analyses using ordinary least squares regression via PROCESS version 3 (ref. ³⁶) in IBM SPSS Statistics version 27 (ref. ¹⁰¹). Performance on the Countermanding task was entered as a parallel mediator. In the next step, each RCT was evaluated separately using the same parallel two-mediator models. The independent variable was training group (*N*-back training or control), the mediators were average post-test performance on untrained *N*-back tasks and average RT on incongruent trials of the Countermanding task, and the dependent variable was post-test performance on Matrix Reasoning. Note that no mediator was modelled as influencing another mediator in the model. To control for pre-existing individual differences in cognitive ability, we entered performance for the three aforementioned tasks as covariates. Since the antecedent variable is an experimental variable in a randomized study, its causal relation to the mediator and the outcome variable can be interpreted unambiguously^{102,103}. Because the post-test data were collected within the same session, causality between the mediator and the outcome variable cannot be established unequivocally; that said, we defend this causal claim on theoretical grounds, which should be sufficient to allow for interpretation of the model³⁶. However, to address this limitation, in the Replication study, Near Transfer performance preceded post-test performance on Matrix Reasoning and other G_T tasks by one day; hence, it represents a plausible causal agent for these outcome measures (see below). Note that in the Results section, s.e. refers to standard error, and $\text{ES}_{\text{p.s.}}$ refers to partially standardized effect size³⁶.

Replication study (RCT 3). *Participants.* A second experiment was conducted at the same universities in spring, summer and fall 2020 to investigate whether the mediation model replicates in a double-blind RCT with two *N*-back training groups and an active control group. The experiment was pre-registered on AsPredicted (no. 40240) and was conducted online due to the COVID-19 pandemic. The participants were randomly assigned to one of three groups: gamified *N*-back training,

non-gamified *N*-back training and general knowledge training (control group), with a target sample size of 64 participants per condition as determined by a power analysis for two-mediator models using Monte Carlo simulations in MPLUS version 8.4 (ref. ¹⁰⁴). We recruited 258 participants, 201 of which completed all training and assessment sessions ($n_{\text{gamified } N\text{-back}} = 70$, $n_{\text{non-gamified } N\text{-back}} = 67$, $n_{\text{control}} = 64$). The participants provided informed consent and were given US\$80 in compensation for participation; all procedures were approved by the University of California, Riverside Institutional Review Board for Human Subjects Research. As can be seen in Supplementary Tables 7 and 8, the sample demographics resemble those reported for the Retrospective study. Unless stated otherwise, the procedures, tasks and training details match those described in the Retrospective study.

Procedure. Since additional tasks were added to be able to examine latent constructs, the experiment consisted of three pre-test and three post-test sessions, with three or four tasks per session to prevent fatigue associated with prolonged testing. Supervision was provided via Zoom for all pre- and post-test sessions, as well as the first training session and any additional sessions as needed, along with technical support via e-mail and phone. Different experimenters supervised pre- and post-test sessions versus training sessions to ensure experimenter blinding; likewise, the participants were not aware that there were multiple training conditions or which one they had been assigned to.

Training tasks. *N*-back training. The same training task was used as in the Retrospective study, and the participants were encouraged to follow the same training protocol—that is, to complete two 20-minute sessions per day over the course of two weeks. The program was set up to prevent access to more than two sessions in 24 hours. The participants were assigned to gamified or non-gamified *N*-back training as described in the Retrospective study. Half of the participants in each *N*-back training group trained on slightly different adaptive algorithms: miniblock-reset (each session starts at 2-back; level up, <3 errors in a 40-trial block; level down, >6 errors in a 40-trial block) and online weighted accuracy (level up, accuracy > 70%; level down, accuracy < 50%), which were collapsed for the purposes of the current analysis.

General knowledge training. Participants in the active control group trained on a task that consists of general knowledge, vocabulary, social science and trivia multiple-choice questions as used previously^{83,105}. They were encouraged to follow the same schedule as the *N*-back training group (two 20-minute sessions per day). This task was selected for the control intervention because it taps into crystallized intelligence and should not engage fluid intelligence or WM. Previous research has shown that participants find this task engaging and enjoyable^{81,83}.

Transfer measures. The first and the last assessment sessions, which consisted of surveys and ETS and DAT tests, were administered via Qualtrics (Qualtrics), whereas the rest were implemented within Recollect the Study. All assessment sessions were supervised via videoconferencing by a researcher who was blind to the training condition the participant had been assigned to. Alternate versions⁵¹ of the ETS Surface Development Test, the ETS Form Board Test and DAT Space Relations were grouped (As and Bs together) for administration at pre- and post-test, the order of which was counterbalanced across participants. Alternate versions of Near Transfer *N*-back, Running Span and Matrix Reasoning were administered at pre- and post-test in a counterbalanced order (A–B, B–A).

Near Transfer *N*-back. The same Near Transfer *N*-back task was used as in the Retrospective study, except that the vehicle stimulus set was replaced with pictures of fruit. The main dependent variable, average accuracy across the 2-back and 3-back levels at pre-test, was comparable across experiments for the two stimulus sets (Retrospective study—vehicles: Mean_{pre} = 0.57, s.d._{pre} = 0.16; Replication study—fruit: Mean_{pre} = 0.60, s.d._{pre} = 0.14).

Running Span. Running Span was an updating WM task in which the participants saw an unpredictable number of stimuli and were asked to report the last *N* items. The same stimuli were used as in Near Transfer *N*-back (animals and fruit), but with different stimulus sets presented at pre- and post-test across the two tasks. The stimuli were presented for 500 ms with an ISI of 1,500 ms. Before each trial, the participants were told how many items they would need to report. The response screen featured a 'keyboard' of pictures of eight stimuli (always shown at the same locations), and the participants were asked to select the last *N* pictures in the order of appearance. The task started with two practice trials (set sizes 2 and 3), and if at least one trial was incorrect, practice was repeated. Feedback was provided during practice but not during the assessment. The assessment consisted of set sizes 2–4 (in that order) with three trials per set size. Progression to set size 5 (and beyond) was contingent on having at least two out of three trials correct on the previous set size. The main dependent measure was the sum of all correctly recalled items (partial score), whereby each item correctly chosen in the correct serial position was assigned one point¹⁰⁶.

Sequencing. In the Sequencing task, the participants were presented with a mixed sequence of letters and numbers that appeared one by one (for example, '8, T, 2, D') and were asked to remember and enter the numbers in numerical order

(for example, '2, 8') followed by the letters in alphabetical order (for example, 'D, T')⁶³. The stimuli appeared for 1,000 ms with an ISI of 500 ms. After five practice trials, the task started at the lowest set size of two items, and then two trials were presented per set size until both trials on a set size were incorrect. The main dependent variable was the sum of all correctly recalled trials (score).

Simple and Complex spatial span. Simple Span was used to assess spatial WM storage, while Complex Span was used to measure WM storage and processing¹⁰⁷. In Simple Span, a sequence of gophers appeared in 12 possible locations, and the participants were tasked with reproducing the sequence by tapping on the locations. The stimuli appeared for 1,500 ms with an ISI of 500 ms. In Complex Span, the same procedure was adopted, except that between each appearance of a gopher, the participant completed a secondary sorting task (dragging an item to the left or right). This secondary task presents interleaved processing demands that are thought to limit the use of mnemonic strategies, such as chunking. In both tasks, the set size (that is, the number of gopher locations in the sequence) ranged from 2 to 10; the test started with the lowest set size and then increased (or decreased) using an adaptive algorithm. The main dependent measure for each task was the sum of all correctly recalled items (partial-credit scoring¹⁰⁸).

Countermanding. The same Countermanding task was used as in the Retrospective study.

UCancellation. UCancellation is a timed test of selective attention and concentration that has been validated against the D2 Test of Attention^{109,110}. The participants were presented with a series of rows featuring pictures of dogs or monkeys, some of which had inverted colours and were reflected over the vertical axis or presented upside down. There were two types of targets, an upright dog and an upside-down monkey, and there were three to five of these targets per row. The goal was to select all targets from left to right within a six-second time limit. Each target type was practised separately in two single blocks prior to proceeding to a mixed block. If a participant cleared a row before the six-second time limit, they could press a button to continue to the next row; thus, participants completed bonus rows if a global time limit of three minutes and ten seconds was not exceeded. The main dependent measure was concentration performance ($\sum \text{Hits} - \sum \text{False alarms}$) in the mixed block.

Matrix Reasoning. The same Matrix Reasoning task, UCMRT, was used as in RCT 2 (ref. ⁷⁴).

ETS Surface Development Test. Surface Development involves visualizing how a piece of paper could be folded to form an object^{83,111,112}. Two images were presented on a computer screen: an unfolded piece of paper and an object. The participants were then asked to mark which edges went together. One practice problem was followed by six test problems, with a time limit of six minutes. The dependent variable was the proportion of correct responses within the time limit.

ETS Form Board Test. In the Form Board test, a given trial consisted of a set of five two-dimensional shapes that could be combined into a two-dimensional geometrical shape presented above the set^{83,112,113}. The goal was to indicate which of the pieces could be combined to match the geometrical shape given at the top of the page. The participants were given two practice problems and then eight minutes to solve 24 problems. The dependent variable was the proportion of correctly solved items in the given time limit.

DAT Space Relations. In the Space Relations test, the participants were presented with a target pattern and four possible objects and were asked to select the appropriate object into which the pattern could be folded^{83,112,114}. Two practice problems preceded 17 test items, with a five-minute time limit to complete the test. The dependent variable was the number of correctly solved items in the given time limit.

Outliers. Of the 201 participants who completed the study, 6 were removed due to administration error on the ETS and DAT tests. Outliers were removed if a given data point was three or more standard deviations from the mean at pre- or post-test. No outliers were removed for Near Transfer or Matrix Reasoning, and six outliers were removed on the basis of Countermanding speed; hence, the resulting sample size was 189 (128 in the *N*-back training group and 61 in the active control group) (Table 3). This dataset was subject to initial replication mediation analyses. In the next step, additional outliers were removed for Running Span ($n = 8$) as well as 11 participants who showed non-compliance on the Sequencing and Spatial Span tasks. This second analytical dataset, which consisted of 170 participants (115 in the *N*-back training group and 55 in the active control group), was used for the additional mediation analyses that included additional assessments and composite scores.

Composite variables. Pre-test correlations of performance on the test battery revealed significant correlations among all measures (Supplementary Table 5) except for Running Span, which correlated only with Sequencing. Moreover,

Running Span showed the lowest test–retest reliability among measures, as indicated by the pre–post correlation observed in the active control group ($r = 0.35$); it was therefore excluded from subsequent analyses. Principal component analysis with oblimin rotation was used to identify composite scores. The sample size for principal component analysis was adequate (Kaiser–Meyer–Olkin index = 0.79), Bartlett’s test of sphericity was significant ($\chi^2(45) = 387.55$, $P < 0.001$), the diagonals of the anti-image correlation matrix were all over 0.7 and the communalities were all above 0.3, thereby confirming that each measure shared some common variance with other measures. A three-factor solution was obtained, which explained 58.5% of the variance (Supplementary Table 9). The following factors were identified: IC (Countermanding and UCancellation), reasoning ability (UCMRT, ETS Surface Development Test, ETS Form Board Test and DAT Space Relations) and WM (Near Transfer N-back, Sequencing, Simple Span and Complex Span), which is in line with the theoretical structure proposed during pre-registration. Composite variables were created by taking the mean of z-transformed dependent variables that had their primary loadings on each factor. Note that for models in which Near Transfer was used as a separate mediator, the WM composite consisted only of Sequencing and Simple and Complex spatial span tasks. For the IC composite, the values on Countermanding were reversed so that higher value indicates better performance.

Reporting summary. Further information on research design is available in the Nature Research Reporting Summary linked to this article.

Data availability

The data that support the findings of this study are available from the corresponding author upon request.

Code availability

Recollect the Study is a free app for Apple and Android devices; however, the experimental code is not open source. A collection of useful Unity tools and scripts developed at the UC Riverside Brain Game Center is available at https://github.com/UCRBrainGameCenter/BGC_Tools. Analysis code (SPSS script) is available from the corresponding author upon request.

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Author contributions

All authors designed the experiments and monitored the data collection. A.P. analysed the data and wrote the manuscript with input from A.R.S. and S.M.J. Senior authorship is shared by A.R.S. and S.M.J.

Competing interests

S.M.J. has an indirect financial conflict of interest with the MIND Research Institute, whose interests are related to this work. The remaining authors declare no competing interests.

Additional information

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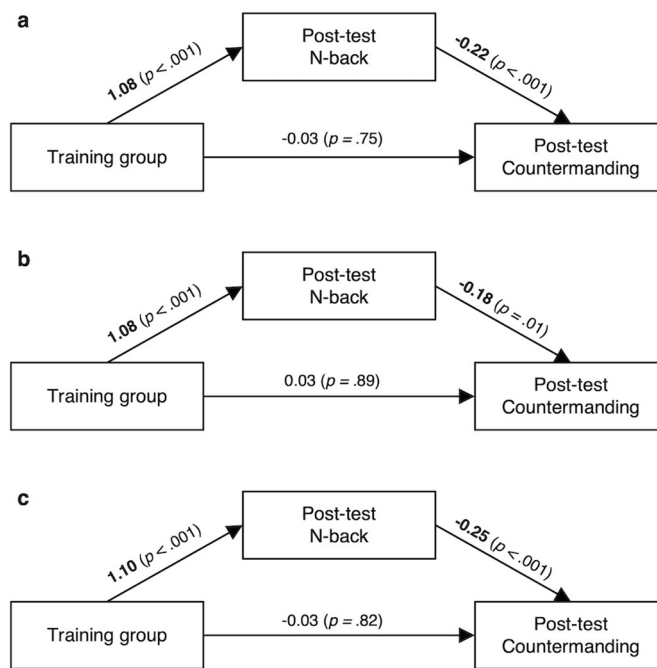
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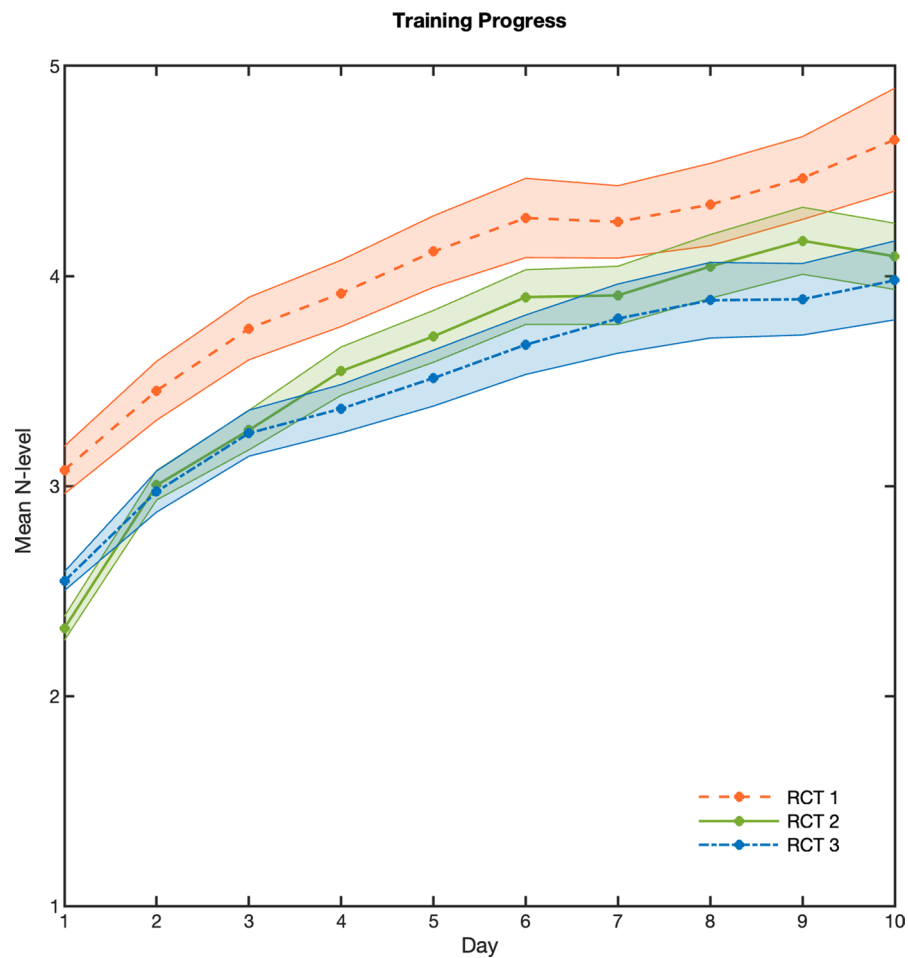
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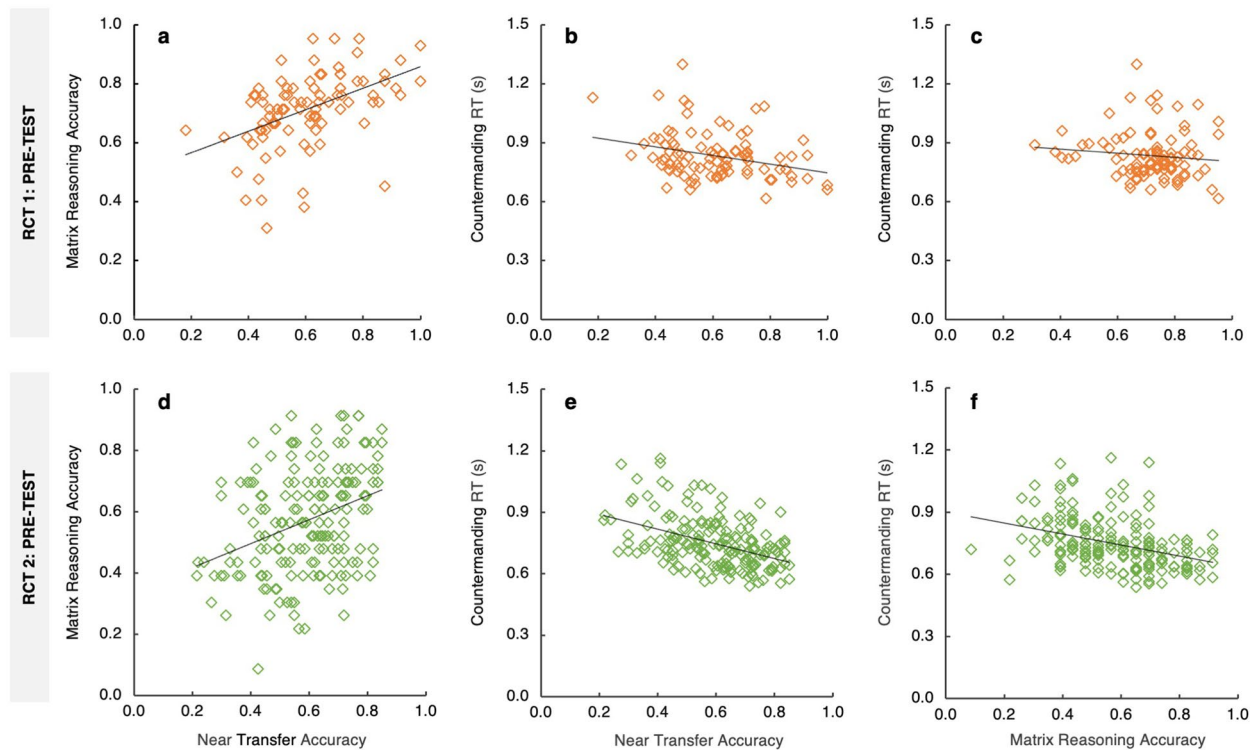
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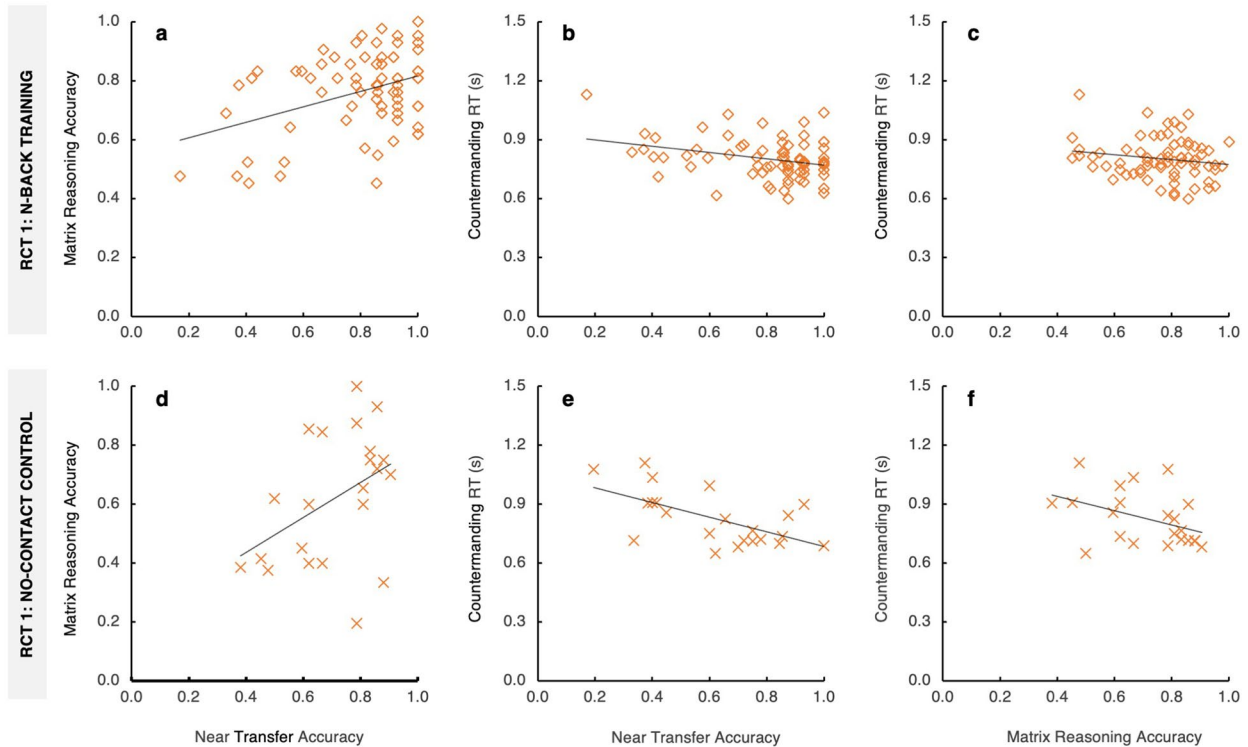
Extended Data Fig. 1 | Schematic of single mediator models with post-test Countermanding as the dependent variable. Path coefficients are standardized regression coefficients with p values reported in brackets. Bold text indicates statistical significance. Covariates (not displayed): pre-test Countermanding and pre-test N-back. **a**, Retrospective study, whole sample ($N = 271$). **b**, Retrospective study, RCT 1 ($N = 94$). **c**, Retrospective study, RCT 2 ($N = 177$).



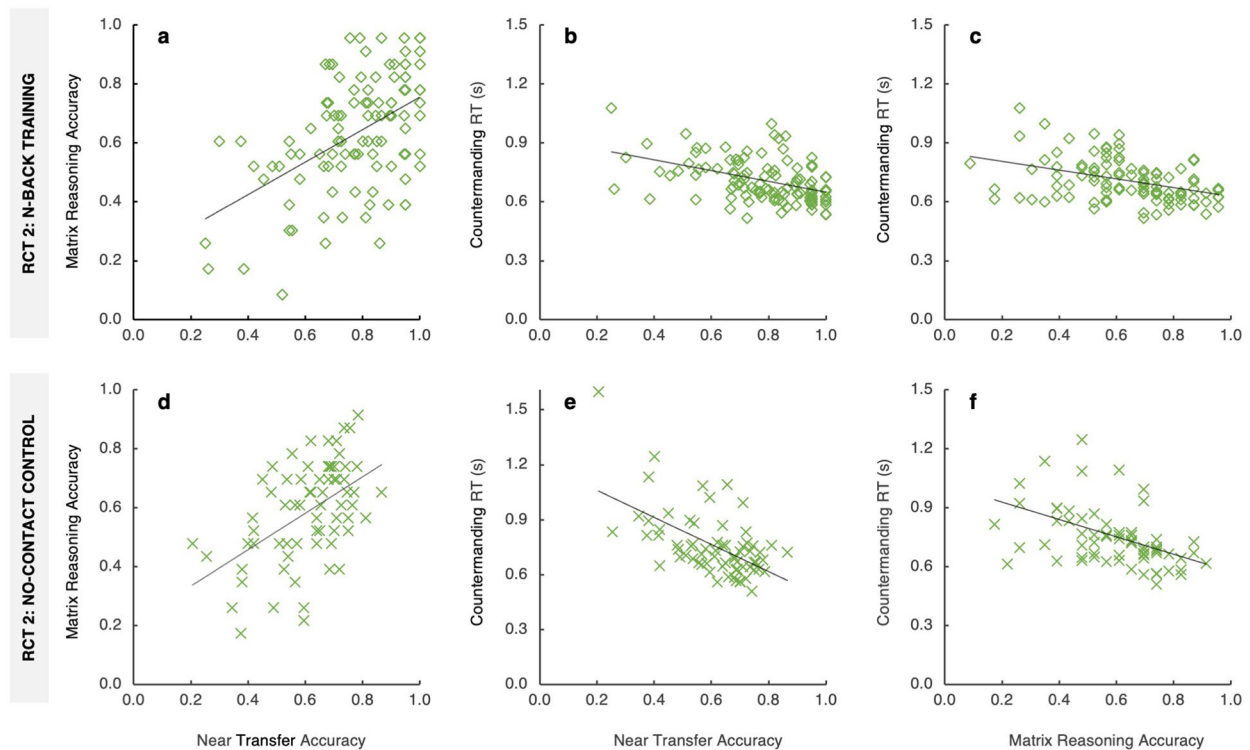
Extended Data Fig. 2 | N-back training progress. Mean N-level achieved per day (40 minutes of training) for active groups in RCT 1 (N = 72), RCT 2 (N = 109), and RCT 3 (N = 128), weighted by the number of trials. RCT differences may reflect the use of slightly different adaptive algorithms. Shaded areas represent standard error of the mean.



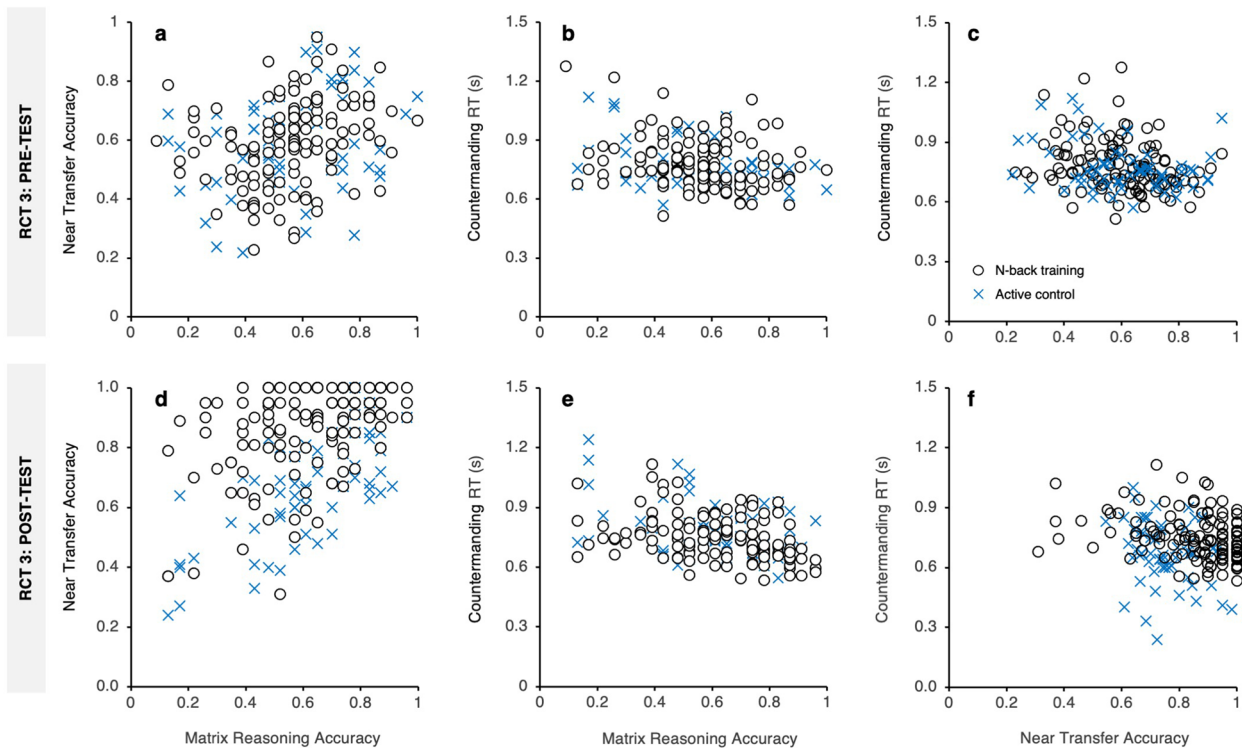
Extended Data Fig. 3 | Scatter plots demonstrating relationships between main outcome measures at pre-test in RCT 1 (N = 94) and RCT 2 (N = 177). Data is collapsed across training conditions and solid lines represent the linear fit of data. **a**, Near Transfer (untrained N-back) and Matrix Reasoning in RCT 1. **b**, Near Transfer and Countermanding in RCT 1. **c**, Matrix Reasoning and Countermanding in RCT 1. **d**, Near Transfer and Matrix Reasoning in RCT 2. **e**, Near Transfer and Countermanding in RCT 2. **f**, Matrix Reasoning and Countermanding in RCT 2.



Extended Data Fig. 4 | Scatter plots demonstrating relationships between main outcome measures at post-test in RCT 1. Solid lines represent the linear fit of data. **a**, Near Transfer (untrained N-back) and Matrix Reasoning performance for N-back training group ($N=72$). **b**, Near Transfer and Countermanding performance for N-back training group ($N=72$). **c**, Matrix Reasoning and Countermanding performance for N-back training group ($N=72$). **d**, Near Transfer and Matrix Reasoning performance for No-contact control group ($N=22$). **e**, Near Transfer and Countermanding performance for No-contact control group ($N=22$). **f**, Matrix Reasoning and Countermanding performance for No-contact control group ($N=22$).



Extended Data Fig. 5 | Scatter plots demonstrating relationships between main outcome measures at post-test in RCT 2. Solid lines represent the linear fit of data. **a**, Near Transfer (untrained N-back) and Matrix Reasoning performance for N-back training group (N=109). **b**, Near Transfer and Countermanding performance for N-back training group (N=109). **c**, Matrix Reasoning and Countermanding performance for N-back training group (N=109). **d**, Near Transfer and Matrix Reasoning performance for No-contact control group (N=68). **e**, Near Transfer and Countermanding performance for No-contact control group (N=68). **f**, Matrix Reasoning and Countermanding performance for No-contact control group (N=68).



Extended Data Fig. 6 | Scatter plots demonstrating relationships between main outcome measures at pre-test and post-test in RCT 3 (N = 189).

Circles represent the N-back training group (N = 128) and cross marks represent the Active control group (N = 61). **a**, Matrix Reasoning and Near Transfer (untrained N-back) performance at pre-test **b**, Matrix Reasoning and Countermanding performance at pre-test. **c**, Near Transfer and Countermanding performance at pre-test. **d**, Matrix Reasoning and Near Transfer performance at post-test **e**, Matrix Reasoning and Countermanding performance at post-test **f**, Near Transfer and Countermanding performance at post-test.

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Study description	Quantitative experimental
Research sample	Our sample consisted of students from the University of California Riverside and Irvine (Retrospective study: N = 283, mean age = 20.1 years, female = 60.9%; Replication study: N = 201, mean age = 20.5 years, female = 78.4%) that completed all assessment and training sessions in the study. The sample was relatively diverse as shown in Supplemental table 8 but because of the narrow age range and geographical location it is not representative of the general population.
Sampling strategy	We used a convenience sampling procedure. Sample size for the Retrospective study was determined using G*Power 3.1.7108: considering $\alpha = .05$, $1-\beta = .8$ and $f = .15$ (which corresponds to $d = .30$ as observed in the literature), we calculated a total sample size of N=35 per group, recruiting an additional 20% to account of for attrition. Within each cohort, and within each quarter of data collection, participants were randomly assigned to one of 3 or 4 possible groups. Sample size for the Replication study was estimated based on a power analysis for two-mediator models using Monte Carlo simulations in MPLUS. Population values for each parameter were obtained from a similar WM training study conducted at our lab. Results indicated that a total sample size of 190 would be sufficient to detect a significant indirect effect through an untrained N-back task with at least .81 probability to reject a false null hypothesis. Hence our target sample size per condition was 64 with a stopping rule of 50 or more based upon available enrollments during the COVID-19 pandemic.
Data collection	The Retrospective study was single-blinded and took place in a laboratory. The Replication study was double-blinded and was conducted remotely due to the COVID-19 pandemic. Data collection involved tablet computers, computers, and mobile devices. The order of assessments and training sessions was coded in advance, and the tasks were equipped with instructions and performance-gating practice trials, therefore minimal involvement from the research assistant was required during the session.
Timing	Retrospective study RCT 1: Winter 2017 - Spring 2017; Retrospective study RCT 2: Fall 2017 - Spring 2018; Replication study RCT 3: Spring 2020 - Fall 2020.
Data exclusions	Retrospective study: Six outliers were removed from the sample based on performance on near transfer and far transfer ($z > 3$ at pre- or post-test separately for each cohort). In the active group, 6 additional participants were removed due to a software error in the training program. The final analytical sample in the Retrospective study consisted of 271 participants. Replication study: Out of 201 participants who completed the study, 6 participants were removed due to administration error on ETS and DAT tests. Outliers were removed if a given data point was 3 or more standard deviations from the mean at pre- or post-test. Six outliers were removed based on Countermanding speed; hence the resulting analytical sample size in the Replication study was 189.
Non-participation	A 75% completion rate was observed in the Retrospective study and a 78% completion rate in the Replication Study. Since the studies consisted of a large number of sessions, the most common reason to drop out was lack of time. Note that participants were not required to give a reason to drop out.
Randomization	Random assignment to experimental groups.

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Population characteristics	See above
Recruitment	Students were recruited to participate in the study via flyers, e-mails, and word of mouth ("We are currently recruiting subjects for a Video Game Training Study / Study on Assessment and Training of Cognitive Function. This study involves playing a game on an electronic device to assess its impact on cognitive abilities").
Ethics oversight	University of California Riverside, IRB Socio-Behavioral with University of California, Irvine IRB reliant on UC Riverside.

Note that full information on the approval of the study protocol must also be provided in the manuscript.